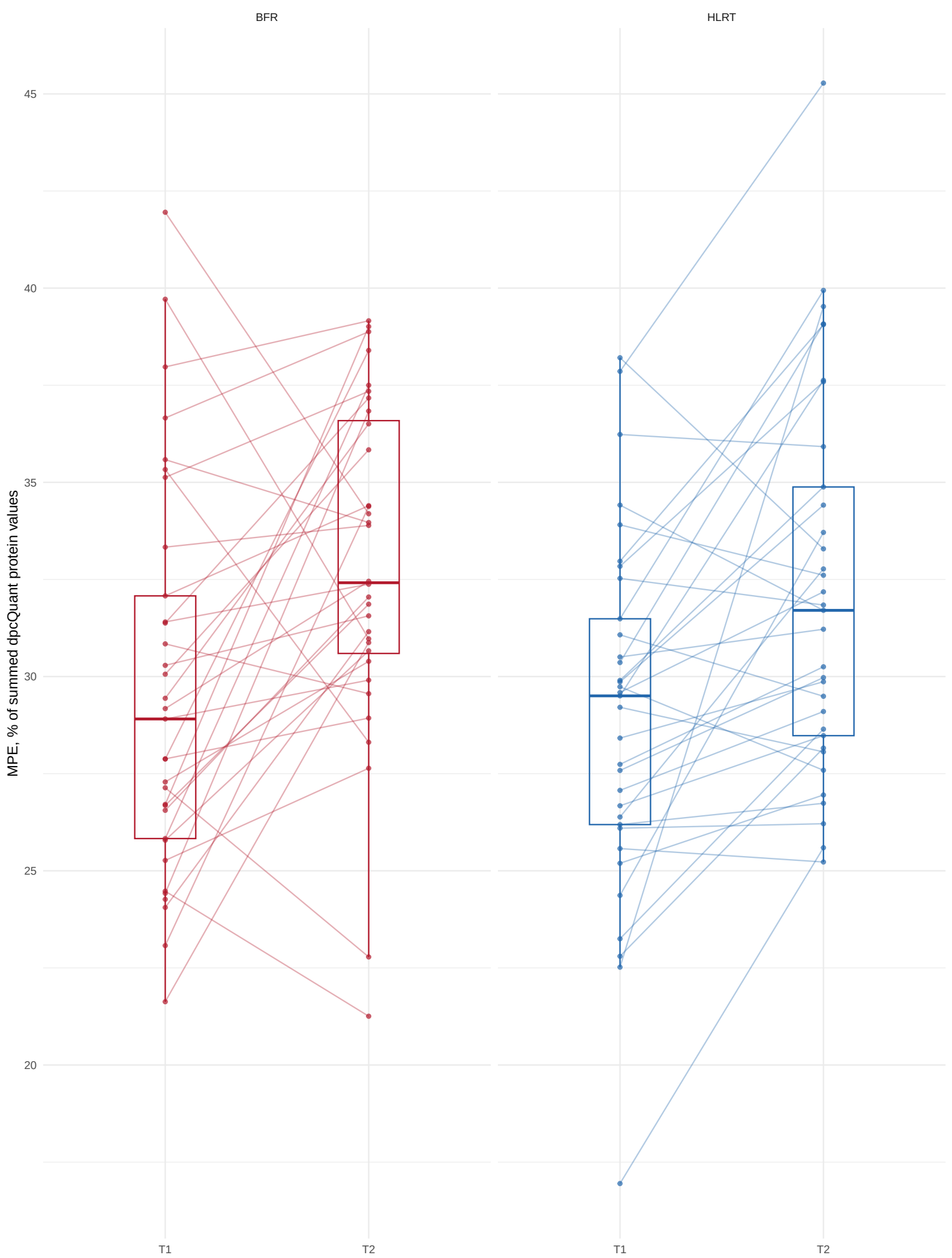
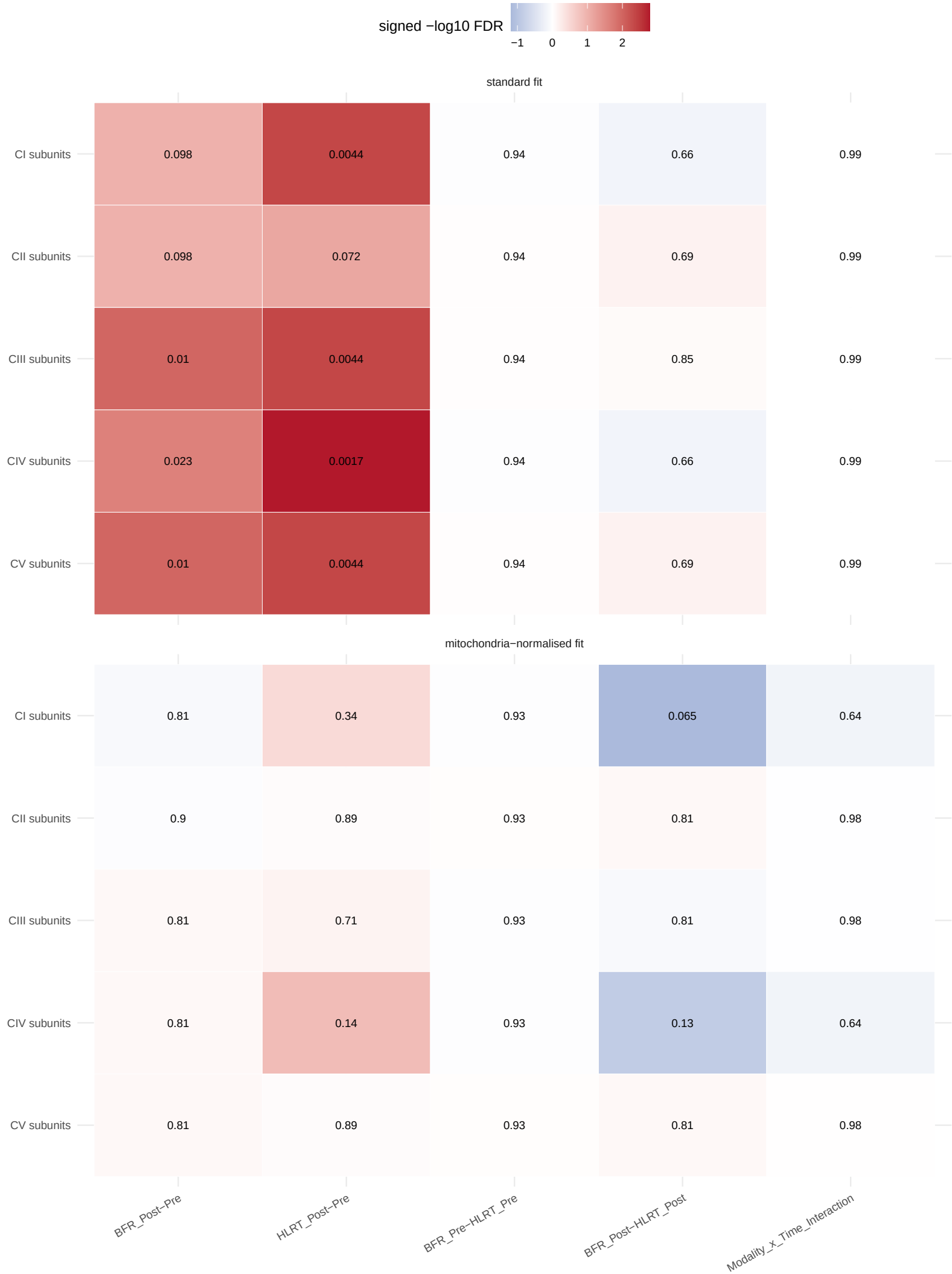
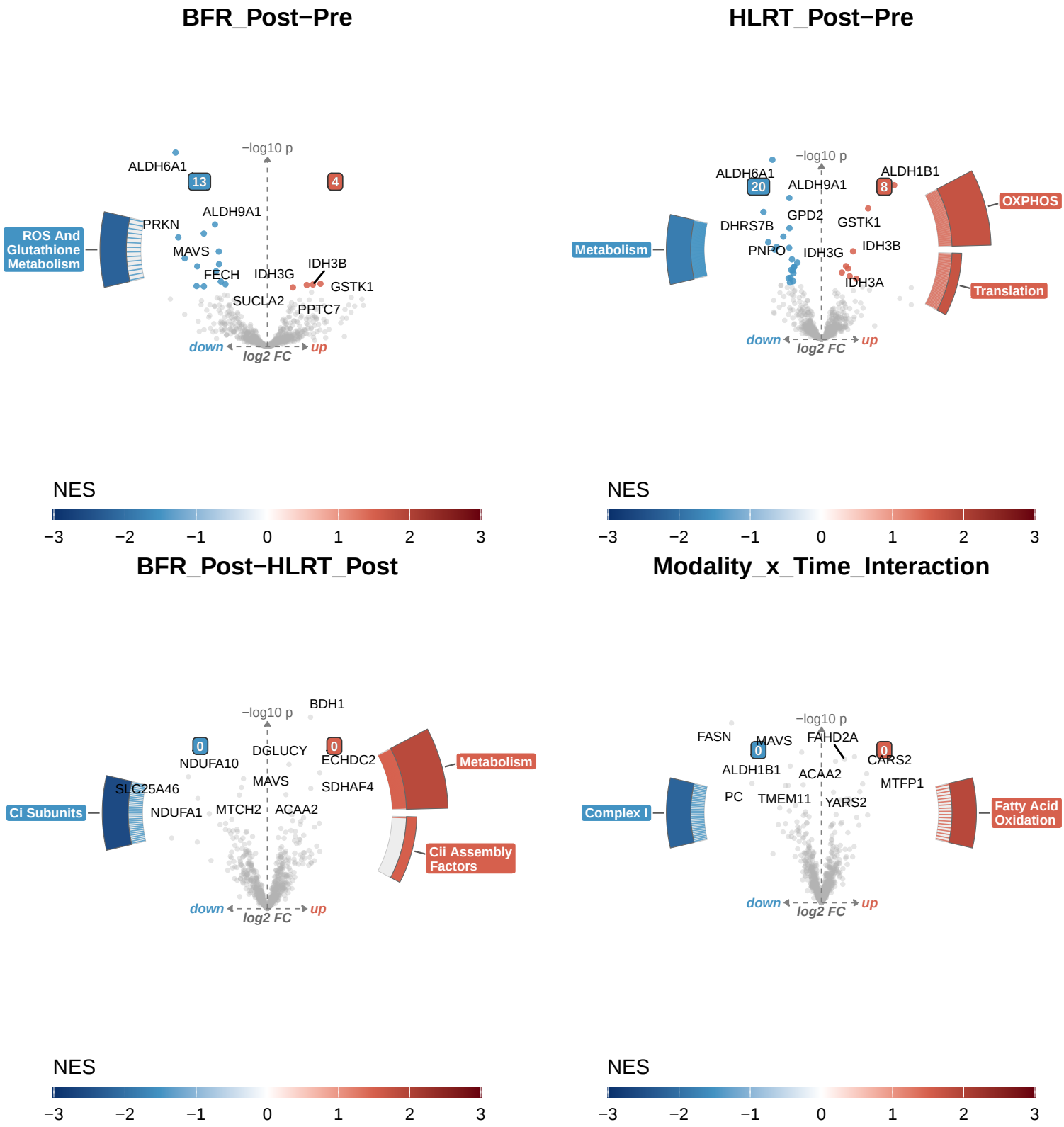


Mitochondrial protein enrichment by arm and time. MPE is the summed dpcQuant value of the 660 measured MitoCarta 3.0 proteins over that of all 3,042 proteins, per biopsy, before normalisation. dpcQuant puts each protein at its typical precursor intensity, so each protein counts once. Lines join each leg before and after training. Tests: mpe_test sheet of 02_test_mito.xlsx.



Complexes I to V, standard and mitochondria−normalised fits. fry on the structural subunits of each complex; labels give fry FDR within the contrast, colour its direction. The standard fit asks whether a complex changed; the normalised fit whether it changed beyond overall mitochondrial content. Data: complex_tests sheet.



Mitochondria−normalised proteins. Points are the 660 MitoCarta proteins after normalising them alone, coloured at BH FDR < 0.05, five named per direction. The ring shows up to twelve MitoPathways at fgsea FDR < 0.05, one per overlapping cluster, coloured by NES; ticks mark leading−edge proteins. fgsea treats proteins as independent, so ringed pathways that fry does not call are leads. Data: mito_proteins and pathway_tests sheets.